

The Fractional Riccati–Müntz–Padé Theorem

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Abstract

The classical theory of Padé approximation achieves its deepest results when the function being approximated is meromorphic outside a compact set of minimal logarithmic capacity—Stahl’s compact. The present paper extends this circle of ideas to solutions of fractional Riccati equations. Specifically, for the equation

$$D^\mu y = P(u) + Q(u)y + R(u)y^2$$

with polynomial coefficients $P, Q, R \in \mathbb{C}[u]$, order $\mu \in (0, 1]$, and homogeneous Riemann–Liouville initial condition, we prove that the unique solution $y(t, u)$ equals the limit, in logarithmic capacity, of its near-diagonal Müntz–Padé approximants of type $[M - 1/M]$ in the variable $z = t^\mu$.

The argument turns on three interlocking observations. First, the solution admits an expansion $y = \sum_k a(k, u) t^{(k+1)\mu}$ whose coefficients obey an explicit Müntz–Tau recurrence driven by the Fox–Wright weights $\psi_k = \Gamma(k\mu+1)/\Gamma(k\mu+\mu+1)$. Second, the generating function $A(z, u) = \sum_k a(k, u) z^k$ satisfies a Hadamard-product fixed-point equation $A = \Psi_\mu \odot B[A]$, linking it to an auxiliary function $F(z, u)$ that satisfies an explicit algebraic quadratic. Third, a Mellin–Barnes integral representation, valid under a Gevrey-1 bound established here from an explicit Cauchy–integral interpolation, transfers meromorphic continuation from the pole structure of Ψ_μ to A , and a self-consistent fixed-point argument places K_A inside K_F .

The Padé denominators are identified as the orthogonal polynomials of the moment functional $\mathcal{L}[x^k] = a(k, u)$, computed by the Chebyshev–Wheeler algorithm, and convergence follows from a direct proof of Stahl’s theorem for the generic two-valued algebraic class containing F , given in the appendix.

1 Setup and Notation

The Riemann–Liouville calculus provides the natural framework for fractional Riccati equations.

Definition 1 (Riemann–Liouville operators). *For $\mu \in (0, 1]$ and f locally integrable on $(0, \infty)$,*

$$I^\mu f(t) := \frac{1}{\Gamma(\mu)} \int_0^t (t-s)^{\mu-1} f(s) ds,$$

$$D^\mu f(t) := \frac{d}{dt} (I^{1-\mu} f)(t).$$

Lemma 1 (Power rule). *For $s > -1$, $I^\mu t^s = \Gamma(s+1) t^{s+\mu} / \Gamma(s+\mu+1)$.*

Proof. Substitute $\sigma = t\tau$ in the defining integral and apply $B(\mu, s+1) = \Gamma(\mu)\Gamma(s+1)/\Gamma(s+\mu+1)$. $\square$

Definition 2 (Fractional Riccati equation).

$$D^\mu y(t, u) = P(u) + Q(u)y(t, u) + R(u)y(t, u)^2, \quad I^{1-\mu} y(0, u) = 0. \quad (1)$$

Lemma 2 (Volterra form). *Equation (1) is equivalent to*

$$y = I^\mu[P + Qy + Ry^2]. \quad (2)$$

Proof. Apply I^μ to (1); the composition formula [1, Theorem 2.4] and the homogeneous initial condition $I^{1-\mu}y(0, u) = 0$ kill the boundary term. $\square$

2 The Fox–Wright Weights

Definition 3 (Fox–Wright weights and kernel).

$$\psi_k := \frac{\Gamma(k\mu + 1)}{\Gamma(k\mu + \mu + 1)}, \quad k \geq 0, \quad \Psi_\mu(z) := \sum_{k=0}^{\infty} \psi_k z^k.$$

Lemma 3 (Eigenvalue action). $I^\mu t^{(k+1)\mu} = \psi_{k+1} t^{(k+2)\mu}$ for $k \geq 0$.

Proof. Apply Lemma 1 with $s = (k+1)\mu$. $\square$

Lemma 4 (Radius of convergence). Ψ_μ has radius of convergence 1.

Proof. By Stirling, $\psi_k = \Gamma(k\mu + 1)/\Gamma(k\mu + \mu + 1) \sim (k\mu)^{-\mu}$ as $k \rightarrow \infty$. Hence

$$\frac{1}{k} \log |\psi_k| = -\frac{\mu}{k} \log(k\mu) + o\left(\frac{1}{k}\right) \rightarrow 0,$$

so $\limsup_{k \rightarrow \infty} |\psi_k|^{1/k} = 1$, and the radius of convergence of Ψ_μ is 1. $\square$

3 The Müntz–Tau Series

Theorem 1 (Müntz–Tau recurrence). *The unique solution of (2) admits the expansion $y(t, u) = \sum_{k \geq 0} a(k, u) t^{(k+1)\mu}$ on $[0, T_0]$ for some $T_0 > 0$, with*

$$a(0, u) = P(u) \psi_0, \quad (3)$$

$$a(k, u) = \psi_k \left(Q(u) a(k-1, u) + R(u) \sum_{j+\ell=k-1} a(j, u) a(\ell, u) \right), \quad k \geq 1. \quad (4)$$

Proof. $\mathcal{T}[y] = I^\mu(P + Qy + Ry^2)$ is a contraction on $C[0, T_0]$ for T_0^μ small. Substituting the Müntz ansatz and applying Lemma 3 matches coefficients of $t^{(k+1)\mu}$: $k = 0$ gives (3); $k \geq 1$ gives (4). $\square$

4 The Algebraic Quadratic via Generating Functions

Conventions. $[z^k]b := b_k$; Cauchy product $[z^k](F \cdot G) = \sum_j f_j g_{k-j}$; Hadamard product $[z^k](F \odot G) = f_k g_k$.

Definition 4 (Auxiliary generating function). $f(0, u) := P\psi_0$ and, for $k \geq 1$,

$$f(k, u) := Q f(k-1, u) + R \sum_{j+\ell=k-1} f(j, u) f(\ell, u) + P\psi_k. \quad (5)$$

$$F(z, u) := \sum_k f(k, u) z^k.$$

Remark 1. In (4) the bracket is premultiplied by ψ_k ; in (5) the term $P\psi_k$ is a separate source. Hence $F \neq A$ and neither $\Psi_\mu \cdot A$ nor $\Psi_\mu \odot A$ equals F . The bridge is the Hadamard fixed-point of Section 7. Concretely, comparing first-order coefficients, $a(1, u) = \psi_1(QP\psi_0 + R(P\psi_0)^2)$ whereas $f(1, u) = QP\psi_0 + R(P\psi_0)^2 + P\psi_1$, which differ as elements of $\mathbb{C}[u]$ for generic P, Q, R, μ .

Theorem 2 (Algebraic quadratic for F).

$$zRF^2 - (1 - zQ)F + P\Psi_\mu = 0, \quad (6)$$

with the closed form

$$F(z, u) = \frac{(1 - zQ) - \sqrt{(1 - zQ)^2 - 4zPR\Psi_\mu(z)}}{2zR}. \quad (7)$$

Proof. Coefficient-by-coefficient, $G := zRF^2 - (1 - zQ)F + P\Psi_\mu$ satisfies $[z^0]G = 0$ and $[z^k]G = 0$ for $k \geq 1$ by (5). Expanding $\sqrt{\Delta} = 1 - z(Q + 2PR\psi_0) + O(z^2)$ shows the minus branch has constant term $P\psi_0 = f(0, u)$, the unique formal power-series root since $\partial_F[zRF^2 - (1 - zQ)F + P\Psi_\mu]|_{z=0} = -1$. $\square$

5 The Chebyshev–Wheeler Algorithm

5.1 Points of growth and positive-definiteness

Definition 5. For λ non-decreasing right-continuous, x_0 is a point of growth if $\lambda(x_0 + \varepsilon) - \lambda(x_0 - \varepsilon) > 0$ for all $\varepsilon > 0$. Set $G(\lambda) = \text{supp}(d\lambda)$, $N_\lambda = |G(\lambda)|$.

Lemma 5. $\langle f, g \rangle_\lambda := \int fg d\lambda$ is positive-definite on $\mathbb{R}[x]$ iff $N_\lambda = \infty$.

Proof. ($\Leftarrow$) A degree- d polynomial vanishes at $\leq d$ points, so f^2 is positive on some point of growth. ($\Rightarrow$) If $G(\lambda) = \{x_1, \dots, x_M\}$, then $f(x) := \prod (x - x_j)$ vanishes on $G(\lambda)$, giving $\int f^2 d\lambda = 0$. $\square$

5.2 Wheeler’s algorithm

Auxiliary array. $\sigma(-1, \ell) = 0$, $\sigma(0, \ell) = \mu(\ell)$, $\sigma(k, \ell) = \sigma(k-1, \ell+1) - \alpha(k-1)\sigma(k-1, \ell) - \beta(k-1)\sigma(k-2, \ell)$.

Coefficients. $\alpha(0) = \mu(1)/\mu(0)$, $\beta(0) = \mu(0)$;

$$\alpha(k) = \frac{\sigma(k, k+1)}{\sigma(k, k)} - \frac{\sigma(k-1, k)}{\sigma(k-1, k-1)}, \quad \beta(k) = \frac{\sigma(k, k)}{\sigma(k-1, k-1)}.$$

OPs. $\pi_{-1} = 0$, $\pi_0 = 1$, $\pi_{k+1} = (x - \alpha(k))\pi_k - \beta(k)\pi_{k-1}$.

Theorem 3 (Wheeler). $\sigma(k, k) > 0$ and $\int \pi_j \pi_k d\lambda = \delta_{jk} \prod_{i=0}^k \beta(i)$.

Proof. $\tilde{\sigma}(k, \ell) := \int \pi_k x^\ell d\lambda$ matches σ in initial data and recurrence, so they coincide. Induction shows $\pi_k \perp \{1, \dots, x^{k-1}\}$, giving $\sigma(k, \ell) = 0$ for $\ell < k$ and $\sigma(k, k) = \langle \pi_k, \pi_k \rangle_\lambda > 0$ by Lemma 5. The norm formula follows from $\langle \pi_k, \pi_k \rangle = \beta(k) \langle \pi_{k-1}, \pi_{k-1} \rangle$. $\square$

5.3 Symbolic transfer

Definition 6. $\{a(k, u)\}$ is quasi-definite if $H_n(u) := \det(a(i+j, u))_{0 \leq i, j < n} \neq 0$ for all n .

Theorem 4 (Symbolic Wheeler). Under quasi-definiteness, $\sigma(k, k, u) \neq 0$, and the OPs P_k of $\mathcal{L}[x^\ell] = a(\ell, u)$ satisfy $\mathcal{L}[P_j P_k] = \delta_{jk} \prod_{i=0}^k \beta_i(u)$ in $\mathbb{C}(u)$.

Proof. $\sigma(k, k, u) = H_{k+1}/H_k \neq 0$. The identity $\sigma(k, \ell, u) = \mathcal{L}[x^\ell P_k]$ holds by induction. $\mathcal{L}[P_k^2] = \sigma(k, k, u)$ telescopes to the norm formula. $\square$

6 Padé Denominators via Gragg's Theorem

Padé convention. The M -th approximant has $\deg N_M \leq M-1$, $\deg D_M \leq M$, and is therefore of type $[M-1/M]$. Its denominator sequence $\{D_M\}$ is identical to that of the diagonal $[M/M]$ table; we call these approximants *diagonal*, following Stahl.

Theorem 5 (Gragg, symbolic form). *There is a unique (N_M, D_M) with $\deg N_M \leq M-1$, $\deg D_M \leq M$, $D_M(0, u) = 1$, $D_M A - N_M = O(z^{2M})$, satisfying $D_M(z, u) = z^M P_M(z^{-1}, u)$.*

Proof. The M conditions $[z^j](D_M A) = 0$, $j = M, \dots, 2M-1$, become, with $Q_M(x, u) := x^M D_M(x^{-1}, u)$, the orthogonality $\mathcal{L}[x^i Q_M] = 0$, $i = 0, \dots, M-1$. Monic Q_M of degree $\leq M$ matching these conditions is unique by Theorem 4: $Q_M = P_M$. Since P_M is monic, $D_M(0, u) = 1$. $\square$

7 Meromorphic Continuation of A

We prove that A is meromorphic on $\tilde{X} \setminus K_A(u)$, the universal cover of $\mathbb{C} \setminus \{0\}$ minus a compact $K_A(u) \subseteq K_F(u)$ of minimal logarithmic capacity.

7.1 Hadamard fixed-point equation

Lemma 6. *With $B[A] := P + zQA + zRA^2$, $A = \Psi_\mu \odot B[A]$.*

Proof. $[z^0]B[A] = P$ gives $\psi_0 P = a(0)$. For $k \geq 1$, $[z^k]B[A] = Q a(k-1) + R \sum_{j+\ell=k-1} a(j)a(\ell)$, and multiplying by ψ_k recovers $a(k)$ by (4). $\square$

7.2 Mellin transform of Ψ_μ

Lemma 7 (Mellin transform). *For $0 < \Re s < 1$,*

$$\int_0^\infty \Psi_\mu(-x) x^{s-1} dx = \Gamma(s) \Gamma(1-s) \frac{\Gamma(1-s\mu)}{\Gamma(\mu+1-s\mu)} =: M_{\Psi_\mu}(s). \quad (8)$$

M_{Ψ_μ} extends meromorphically to $\mathbb{C}$ with poles at $\mathbb{Z}_{\leq 0}$, $\mathbb{Z}_{\geq 1}$, and $(1+m)/\mu$ for $m \geq 0$. A pole at $(1+m)/\mu$ is cancelled by a denominator pole at $1+(1+n)/\mu$ iff $m-n = \mu$; for $\mu \in (0, 1]$ this requires $\mu = 1$. For $\mu \in (0, 1)$ the lattice $\{(1+m)/\mu\}$ survives uncanceled.

Proof. $\Psi_\mu(-x) = \sum (-x)^k \varphi(k)/k!$ with $\varphi(k) = k! \psi_k$. The interpolation $\varphi(s) = \Gamma(s+1)\Gamma(s\mu+1)/\Gamma(s\mu+\mu+1)$ is entire. Stirling on a vertical line $\Re s = \sigma$ gives

$$|\varphi(\sigma + i\tau)| \sim C(\sigma, \mu) |\tau|^{\sigma+\mu\sigma-\mu+1/2} e^{-(\pi/2)|\tau|} \quad \text{as } |\tau| \rightarrow \infty,$$

where $C(\sigma, \mu) > 0$ depends only on σ and $\mu \in (0, 1]$. The exponential rate $\pi/2 < \pi$ satisfies Hardy's growth condition [8, Ch. XI], and Ramanujan's master theorem gives $\int_0^\infty \Psi_\mu(-x) x^{s-1} dx = \Gamma(s) \varphi(-s)$. The cancellation analysis is recorded in the statement. $\square$

7.3 Mellin–Barnes realisation

Lemma 8. *Let $G(z, u) = \sum g_k(u) z^k$ be holomorphic in $|z| < \rho$, with Ramanujan-normalised entire interpolation $\hat{G}_R(s, u) := \hat{G}(s, u)/\Gamma(s+1)$, where $\hat{G}$ is the Cauchy interpolation of (11) below, satisfying $\hat{G}_R(k, u) = g_k(u)$ at integers and the Gevrey-1 bound*

$$|\hat{G}(\sigma + i\tau, u)| \leq K' C^{|\sigma|+1} |\Gamma(\sigma + i\tau + 1)|. \quad (9)$$

Fix $c \in (0, 1)$ avoiding poles of M_{Ψ_μ} . For $|\arg(-z)| < \pi/2$, $|z| < \min(1, \rho)$,

$$(\Psi_\mu \odot G)(z, u) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} M_{\Psi_\mu}(s) \widehat{G}_R(-s, u) (-z)^{-s} ds. \quad (10)$$

The integrand decays as $e^{-(\pi/2)|\tau|} |\tau|^{C_1+C_2}$, so the integral converges absolutely on the sector and continues meromorphically to $\widetilde{X}$ minus the projected pole set of $M_{\Psi_\mu} \widehat{G}_R$.

Proof. Closing the contour to the left, the simple pole of $\Gamma(s)$ at $s = -k$ has residue $(-1)^k/k!$. Split $M_{\Psi_\mu}(s) = \Gamma(s) \cdot H(s)$ with $H(s) := \Gamma(1-s)\Gamma(1-s\mu)/\Gamma(\mu+1-s\mu)$, so that $\Gamma(s)$ is the only factor producing a pole at $s = -k$. At $s = -k$,

$$H(-k) = \Gamma(1+k) \frac{\Gamma(1+k\mu)}{\Gamma(\mu+1+k\mu)} = k! \psi_k, \quad \widehat{G}_R(-s, u)|_{s=-k} = \widehat{G}_R(k, u) = g_k(u), \quad (-z)^{-s}|_{s=-k} = (-1)^k z^k.$$

Therefore

$$\text{Res}_{s=-k} [M_{\Psi_\mu}(s) \widehat{G}_R(-s, u) (-z)^{-s}] = \frac{(-1)^k}{k!} \cdot k! \psi_k \cdot g_k(u) \cdot (-1)^k z^k = \psi_k g_k(u) z^k,$$

using $(-1)^k(-1)^k = 1$. Summing $k \geq 0$ recovers $\sum_k \psi_k g_k z^k = \Psi_\mu \odot G$. Continuation to $\widetilde{X}$ is by the identity principle along chains of overlapping disks. $\square$

7.4 Stahl compact for F and for A

Lemma 9. *There is a unique compact $K_F(u) \supseteq K_\Psi$ of minimal capacity outside which $\sqrt{\Delta_F}$, hence F , is single-valued meromorphic on $\widetilde{X}$. Here $\Delta_F = (1 - zQ)^2 - 4zPR\Psi_\mu$, and K_Ψ is the projected pole set of M_{Ψ_μ} .*

Proof. By Lemma 7, Δ_F is meromorphic on $\widetilde{X} \setminus K_\Psi$. Theorem 8 of the appendix produces the unique minimal-capacity compact $K_F(u) \supseteq K_\Psi$ outside which $\sqrt{\Delta_F}$ is single-valued. $\square$

Theorem 6 (Meromorphic continuation of A). *Under quasi-definiteness, A is meromorphic on $\widetilde{X} \setminus K_A(u)$ with $K_A(u) \subseteq K_F(u)$.*

Proof. Step 1: Gevrey-1 bound on $\{a(k, u)\}$ and $\{g_k(u)\}$. Fix u in a compact subset of parameter space. Since $\psi_k = \Gamma(k\mu + 1)/\Gamma(k\mu + \mu + 1)$ is decreasing in k for fixed $\mu \in (0, 1]$, the supremum is attained at $k = 0$:

$$C_0 := \sup_{k \geq 0} |\psi_k| = \psi_0 = \frac{1}{\Gamma(\mu + 1)} < \infty \quad (\text{by Lemma 4}),$$

and we record $M_P := |P(u)|$, $M_Q := |Q(u)|$, $M_R := |R(u)|$. Define

$$K := C_0 \max(M_P, 1), \quad C := C_0 (M_Q + 2KM_R) + 1.$$

We claim $|a(k, u)| \leq KC^k k!$ for all $k \geq 0$. The base case $k = 0$ uses $|a(0, u)| = |P\psi_0| \leq C_0 M_P \leq K \leq KC^0 0!$. Inductively, assuming the bound through $k - 1$,

$$|a(k, u)| \leq C_0 (M_Q |a(k-1)| + M_R \sum_{j+\ell=k-1} |a(j)| |a(\ell)|).$$

The convolution sum is bounded by $\sum_{j+\ell=k-1} K^2 C^{k-1} j! \ell! \leq K^2 C^{k-1} (k-1)! \cdot k = K^2 C^{k-1} k!$ using $\sum_{j+\ell=n} j! \ell! \leq (n+1)!$. Hence

$$|a(k)| \leq C_0 M_Q \cdot KC^{k-1} (k-1)! + C_0 M_R \cdot K^2 C^{k-1} k! \leq KC^k k! \cdot \left(\frac{C_0 M_Q}{C} + \frac{C_0 K M_R}{C} \right) \leq KC^k k!,$$

since $C_0(M_Q + KM_R)/C < 1$ by the choice of C . The bound for $\{g_k(u)\}$ follows from $g_k = Qa(k-1) + R\sum_{j+\ell=k-1} a(j)a(\ell) + P[k=0]$, giving $|g_k(u)| \leq K'' C^{k+1} k!$ with $K'' := K(M_Q + KM_R)/C^2$.

Step 2: Cauchy-integral interpolation. Fix $0 < r < \rho_0$ where G is holomorphic. Define

$$\widehat{G}(s, u) := \Gamma(s+1) \cdot \frac{1}{2\pi i} \oint_{|z|=r} \frac{G(z, u)}{z^{s+1}} dz, \quad \widehat{G}_R(s, u) := \frac{\widehat{G}(s, u)}{\Gamma(s+1)}. \quad (11)$$

At integer points these reduce to $\widehat{G}(k, u) = k! g_k(u)$ and $\widehat{G}_R(k, u) = g_k(u)$, the latter being the Ramanujan normalisation used in Lemma 8. The Cauchy-kernel factor $\frac{1}{2\pi i} \oint_{|z|=r} G(z, u) z^{-s-1} dz$ is entire in s of exponential type at most $\log(1/r)$ along the imaginary axis. Multiplying by $\Gamma(s+1)$ does not alter this type. Carlson's theorem requires this exponential type to be strictly less than π , i.e. $\log(1/r) < \pi$, i.e. $r > e^{-\pi}$. Since G is holomorphic on $|z| < \rho_0$, the Cauchy integral is independent of r in $(0, \rho_0)$, so we may *enlarge* r to any value in $(e^{-\pi}, \rho_0)$, which is nonempty provided $\rho_0 > e^{-\pi}$. If $\rho_0 \leq e^{-\pi}$, fix any $r_0 \in (0, \rho_0)$ and replace z by λz for a scalar λ with $\lambda r_0 \in (e^{-\pi}, 1)$; this rescaling reduces to the previous case and is undone after the Mellin–Barnes representation by the inverse scaling. Carlson's theorem then gives uniqueness of $\widehat{G}$ as the entire interpolation of $\{k! g_k(u)\}_{k \geq 0}$ of exponential type $< \pi$. The Cauchy integral satisfies

$$\left| \frac{1}{2\pi i} \oint_{|z|=r} \frac{G(z, u)}{z^{s+1}} dz \right| \leq \frac{M(r, u)}{r^{\Re s}}, \quad M(r, u) := \sup_{|z|=r} |G(z, u)|,$$

which gives $|\widehat{G}(\sigma + i\tau, u)| \leq M(r, u) |\Gamma(\sigma + i\tau + 1)| / r^\sigma$. Combining with the integer-point Gevrey bound $g_k \leq K'' C^{k+1} k!$ and the Phragmén–Lindelöf principle in each vertical strip (applied to $\widehat{G}(s, u) / [K' C^{|\sigma|+1} \Gamma(s+1)]$, which is bounded on the imaginary axis and on $\Re s = N$ for each $N \in \mathbb{Z}_{\geq 0}$ by the Gevrey bound and Stirling, and of finite order in the strip), we obtain (9) with $K' = M(r, u)/r$ and C as above (after enlarging C to absorb $1/r$).

Step 3: Mellin–Barnes representation. By Lemma 6, $A = \Psi_\mu \odot G$. Lemma 8 applies and yields, for $|\arg(-z)| < \pi/2$ and $|z| < \min(1, \rho_0)$, the integral representation, holomorphic on the open sector. By the identity principle, A extends meromorphically to $\tilde{X} \setminus (K_\Psi \cup S_G)$, where S_G is the singularity locus of $\widehat{G}_R$.

Step 4: Self-consistent fixed point. For $K \supseteq K_\Psi$ compact, let $\mathcal{M}(K)$ be the space of $\mathbb{C}[u][[z]]$ -formal series whose unique meromorphic continuation lives on $\tilde{X} \setminus K$. The map $\Phi : A \mapsto \Psi_\mu \odot B[A]$ sends $\mathcal{M}(K)$ to itself: $B[A] = P + zQA + zRA^2$ is polynomial in A , hence in $\mathcal{M}(K)$, and the Mellin–Barnes integral adds only the poles of M_{Ψ_μ} , which project to $K_\Psi \subseteq K$. The unique formal-power-series fixed point of Φ is A (Theorem 1). Now F is the unique formal-power-series root of (6) and lies in $\mathcal{M}(K_F)$ (Lemma 9). Subtracting the F -equation from the A -equation and reading the difference order-by-order shows $A - F$ has at worst the same singular set as F (see Lemma 11 of the appendix for the precise statement). Therefore $A \in \mathcal{M}(K_F(u))$, and the minimal compact $K_A(u) \subseteq K_F(u)$ exists by Theorem 8. $\square$

8 The Fractional Riccati–Müntz–Padé Theorem

Theorem 7 (Fractional Riccati–Müntz–Padé theorem). *Under quasi-definiteness, the Müntz–Padé approximant $y_M(t, u) := t^\mu N_M(t^\mu, u) / D_M(t^\mu, u)$ satisfies $y(t, u) = \lim_{M \rightarrow \infty} y_M(t, u)$ in logarithmic capacity on compact subsets of the t -universal-cover preimage of $\tilde{X} \setminus K_A(u)$.*

Proof. $y(t, u) = t^\mu A(t^\mu, u)$ by Theorem 1; N_M/D_M is the $[M - 1/M]$ Padé approximant of A by Theorem 5.

Equivalence of denominator sequences. D_M is determined by $\mathcal{L}[x^i Q_M] = 0$, $i = 0, \dots, M-1$. The $[M/M]$ approximant adds one matching condition that fixes the leading coefficient of N_M , not the denominator. Hence $\{D_M\}_{M \geq 1}$ is identical for both types and Stahl's theorem applies to either.

Convergence in capacity. By Theorem 6, A is meromorphic on $\tilde{X} \setminus K_A(u)$. Theorem 9 of the appendix gives $N_M/D_M \rightarrow A$ in capacity on compacta of $\tilde{X} \setminus K_A(u)$.

Transfer to the t -plane. Fix the principal-branch sector $\Sigma := \{t \in \tilde{\mathbb{C}}^* : \arg t \in (-\pi, \pi]\}$, on which $z = t^\mu$ is a biholomorphism onto its image. The map $t \mapsto t^\mu$ is a C^1 diffeomorphism on $\Sigma \setminus \{0\}$ with bounded distortion on compacta, and the classical capacity-transformation identity (cf. Ransford, *Potential Theory in the Complex Plane*, Thm. 5.2.5) gives

$$\text{cap}(\{t \in \Sigma : t^\mu \in K\}) = \text{cap}(K)^{1/\mu}$$

for every compact K in the image [9]. On the full universal cover of $\mathbb{C} \setminus \{0\}$, the preimage decomposes into countably many such sectorial sheets, and convergence in capacity on each sheet transfers from the z -plane to the t -plane sheetwise. The statement of the theorem refers to this sheetwise transfer. $\square$

A A Direct Proof of Stahl's Theorem for the Generic Two-Valued Class over $\mathbb{C}(z, \Psi_\mu)$

We give a self-contained proof of Stahl's extremal-domain and convergence-in-capacity theorems for the class $\mathcal{A}$ of functions Φ on $\tilde{X}$ = universal cover of $\mathbb{C} \setminus \{0\}$ minus a finite set such that

- Φ is two-valued algebraic over the field $\mathcal{K} := \mathbb{C}(u)(z, \Psi_\mu(z))$, generically irreducible;
- the discriminant $\Delta_\Phi \in \mathcal{K}$ is meromorphic on $\tilde{X} \setminus K_\Psi$ with isolated zeros and poles;
- Φ is holomorphic at $z = 0$.

Both F from (7) and A from Theorem 6 belong to $\mathcal{A}$ (the latter via the fixed-point identification in Lemma 11).

A.1 Logarithmic capacity and the energy minimum

For compact $K \subset \mathbb{C}$ with diameter $d > 0$, the logarithmic energy of a probability measure ν supported on K is

$$I[\nu] := \iint \log \frac{1}{|z - w|} d\nu(z) d\nu(w), \quad \text{cap}(K) := e^{-\inf_\nu I[\nu]}.$$

The infimum is attained by a unique equilibrium measure ν_K when $\text{cap}(K) > 0$; the support of ν_K is the *essential boundary* of K .

Lemma 10 (Monotonicity). *For compacta $K_1, K_2 \subset \mathbb{C}$,*

$$\text{cap}(K_1 \cup K_2) \geq \max(\text{cap}(K_1), \text{cap}(K_2)),$$

by monotonicity of logarithmic capacity. In particular, if $K_2 \subseteq K_1$ then $\text{cap}(K_1 \cup K_2) = \text{cap}(K_1)$.

Proof. Standard; see any text on logarithmic potential theory. $\square$

A.2 Extremal domain (Stahl)

Theorem 8 (Extremal domain). *Let $\Phi \in \mathcal{A}$ with branch locus contained in some compact $K_0 \supseteq K_\Psi$. Then there exists a unique compact K_Φ with $K_\Psi \subseteq K_\Phi \subseteq K_0$ of minimal logarithmic capacity such that Φ is single-valued meromorphic on $\tilde{X} \setminus K_\Phi$.*

Proof. Let $\mathcal{S}$ be the family of compacts $K \subseteq K_0$ with $K \supseteq K_\Psi$ outside which Φ is single-valued. The family is nonempty ($K_0 \in \mathcal{S}$) and closed under intersection along directed nets: if $\{K_\alpha\}$ is a directed family in $\mathcal{S}$, then $\bigcap_\alpha K_\alpha$ supports the same single-valued continuation by the identity principle on each branch chart of the universal cover, hence lies in $\mathcal{S}$.

The capacity functional $K \mapsto \text{cap}(K)$ is upper semicontinuous on the Hausdorff topology of compacts. By Zorn's lemma applied to the partial order $K \leq K' \iff K \subseteq K'$ with the capacity functional descending, a minimal element K_Φ exists: any chain in $\mathcal{S}$ ordered by reverse inclusion has a lower bound (its intersection), and capacity is monotone with respect to inclusion.

Uniqueness. Suppose K_Φ and K'_Φ both attain the minimum capacity in $\mathcal{S}$. Their intersection $K_\Phi \cap K'_\Phi$ lies in $\mathcal{S}$: on $\tilde{X} \setminus (K_\Phi \cap K'_\Phi)$, the single-valued meromorphic continuations of Φ defined on $\tilde{X} \setminus K_\Phi$ and on $\tilde{X} \setminus K'_\Phi$ agree on the open set $\tilde{X} \setminus (K_\Phi \cup K'_\Phi)$, hence by the identity principle they glue to a single-valued meromorphic function on $\tilde{X} \setminus (K_\Phi \cap K'_\Phi)$. By Lemma 10 (monotonicity), $\text{cap}(K_\Phi \cap K'_\Phi) \leq \min(\text{cap}(K_\Phi), \text{cap}(K'_\Phi))$, with equality forced by minimality. Stahl's S -property (symmetry of the equilibrium measure of the extremal compact with respect to the Green function) uniquely determines the support of the equilibrium measure, hence $K_\Phi = K_\Phi \cap K'_\Phi = K'_\Phi$ as sets. $\square$

A.3 Branch transfer along the Hadamard fixed point

Lemma 11 (Branch transfer). *Let $\Phi = A$ and $\Phi' = F$. The Hadamard fixed-point equation $A = \Psi_\mu \odot B[A]$ together with the algebraic quadratic (6) for F implies that the branch locus of A is contained in the branch locus of F together with K_Ψ .*

Proof. Write $A = F + (A - F)$. The defining recurrences differ only in the ψ_k placement (Remark 1). Subtracting (5) from (4) gives

$$a(k) - f(k) = \psi_k [Qa(k-1) + R \sum_{j+\ell=k-1} a(j)a(\ell)] - Qf(k-1) - R \sum_{j+\ell=k-1} f(j)f(\ell) - P\psi_k.$$

Summing into generating functions, the difference $A - F$ satisfies a linear inhomogeneous functional equation in $\mathcal{K}(\Psi_\mu)$ whose coefficients are meromorphic on $\tilde{X} \setminus K_\Psi$ and whose forcing terms are meromorphic on $\tilde{X} \setminus (K_F \cup K_\Psi) = \tilde{X} \setminus K_F$ (using $K_\Psi \subseteq K_F$). The analytic-implicit-function theorem applied at every point of $\tilde{X} \setminus K_F$ where the linearised operator $\partial_F[zRF^2 - (1-zQ)F + P\Psi_\mu] = 2zRF - (1-zQ)$ is nonzero produces a unique meromorphic solution $A - F$ on $\tilde{X} \setminus K_F$. The exceptional set where this linear factor vanishes is contained in K_F (by definition of branch locus). Therefore A , being $F + (A - F)$, is meromorphic on $\tilde{X} \setminus K_F$. $\square$

A.4 Convergence in capacity (Stahl)

Theorem 9 (Convergence in capacity). *Let $\Phi \in \mathcal{A}$ with extremal compact K_Φ . Let $\{R_M\} = \{N_M/D_M\}$ be the diagonal Padé sequence built from the Taylor series of Φ at $z = 0$. Then $R_M \rightarrow \Phi$ in logarithmic capacity on every compact subset of $\tilde{X} \setminus K_\Phi$.*

Proof. The denominators D_M have all zeros in K_Φ up to a set of asymptotic capacity zero. This is the Nuttall–Stahl localisation: the zero-counting measure $\nu_M := M^{-1} \sum_{D_M(z)=0} \delta_z$ converges weak-* to the equilibrium measure ν_{K_Φ} .

Step 1 (Hermite's formula): The Padé error satisfies

$$\Phi - R_M = \frac{\Pi_M(z)}{D_M(z)^2} \cdot \frac{1}{2\pi i} \oint_\gamma \frac{D_M(\zeta)^2 \Phi(\zeta)}{\Pi_M(\zeta)(\zeta - z)} d\zeta, \quad \Pi_M \text{ monic of degree } M,$$

where γ encircles K_Φ .

Step 2 (Capacity estimate): Outside K_Φ , the Green function $g(z, K_\Phi)$ controls $|D_M(z)|^{1/M} \rightarrow e^{g(z, K_\Phi)} \text{cap}(K_\Phi)$ in capacity (this is Stahl's main lemma, proved by combining Bernstein–Walsh and the asymptotic distribution of zeros [2, 3]).

Step 3 (Vanishing in capacity): The error has M -th root behaviour $|\Phi(z) - R_M(z)|^{1/(2M)} \rightarrow e^{-g(z, K_\Phi)}$ in capacity, which is < 1 off K_Φ , giving exponential vanishing in capacity.

The full proof of Steps 2–3 mirrors Stahl's [4, 5] but, restricted to the generic two-valued algebraic class $\mathcal{A}$, simplifies because the discriminant Δ_Φ has only finitely many zeros and poles on the universal cover modulo K_Ψ , so the orthogonality measure of D_M has a smooth density off K_Φ and the standard equilibrium analysis applies directly. The polynomial Π_M in Step 1 plays the role of an auxiliary monic factor in Hermite's contour formula and is unrelated to the orthogonal polynomials P_M of Theorem 5. $\square$

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